Hybrid ETAS Ranking of Earthquake Hazard in Türkiye

Barış Kısır
Independent Researcher, Türkiye

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Abstract

Seismicity separates into two well-studied statistical regimes: short-scale aftershock clustering, described by Epidemic-Type Aftershock-Sequence (ETAS) point processes, and long-scale magnitude–frequency regularity, described by the Gutenberg–Richter law. This paper specifies a deterministic, catalogue-only forecasting engine that converts these regularities into *relative* regional hazard rankings over Türkiye and neighbouring tectonic provinces (latitudes 34°–43°N, longitudes 24°–46°E) on a 0.5° grid. Four layers are combined: an ETAS spatio-temporal field with a kernel-smoothed background and a fully evaluated triggering kernel; a Gutenberg–Richter layer for completeness, b-value mapping, and magnitude scaling; six bounded seismicity indicators; and a Brownian-Passage-Time renewal layer for M$\geq$6 and M$\geq$7 recurrence. A threshold-dependent composite score blends the layers with magnitude-specific weights and is normalised to a bounded display scale for the M5+, M6+, and M7+ targets. Every equation is stated together with its numerical constants and illustrated with worked examples computed from the implementation itself; two figures are derived directly from the production catalogue and three from the archived forecast snapshots. Because each forecast uses only events at or before its calculation date, a bundled archive of precomputed snapshots supports exact retrospective reconstruction. Scores are within-catalogue relative rankings, never occurrence probabilities, and the system carries standing messaging that it is not a warning system.

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1 Introduction

The Anatolian plate is squeezed westward between the Eurasian plate to the north and the Arabian and African plates to the south and southwest. Its northern boundary, the right-lateral North Anatolian Fault Zone (NAFZ), and its southeastern boundary, the left-lateral East Anatolian Fault Zone (EAFZ), have each hosted destructive magnitude-7 sequences, including the 1999 İzmit earthquake (M_w7.6) on the NAFZ. Dense instrumental coverage since the 2000s records tens of thousands of locatable events per year in the broader region, which makes Türkiye a natural laboratory for statistical earthquake forecasting: the catalogue is large, the tectonics are localised onto major fault systems, and the observational history is long enough to calibrate both clustering and renewal models.

Earthquake forecasting occupies a deliberate middle ground between two extremes. Exact prediction of the time, place, and magnitude of individual shocks is beyond current observational capability. At the other extreme, time-independent hazard maps describe where large events are possible in principle but say nothing about the current state of the system. Operational forecasting, as codified for example by Jordan and co-workers [24] and tested continuously by the Collaboratory for the Study of Earthquake Predictability [25], assigns expected activity rates that evolve as new data arrive and submits them to prospective, like-for-like testing. The engine described here belongs to this tradition but narrows its claims further: it outputs rankings of regions rather than calibrated rates, on the argument that a transparent ordering of signals is more honest than a probability number whose calibration has not been earned.

The recent literature sharpens the case for a strong classical baseline. Neural point-process models evaluated on chronological earthquake forecasting benchmarks have repeatedly failed to outperform a well-specified ETAS model [29, 30], while hybrid convolutional-recurrent architectures are judged precisely by whether they pass the same CSEP test suite as ETAS [31]. The implication is that the marginal value of additional complexity must be demonstrated against ETAS, not assumed. The present engine therefore keeps ETAS at its centre and adds only bounded, interpretable modifiers whose individual contributions can be read off the composite score term by term.

Three design decisions organise the work:

(D1) Catalogue-only computation. No waveform, geodetic, or learned model state enters the evaluation; every score is a deterministic function of the merged instrumental and historical catalogue and a reference timestamp.

(D2) Threshold-dependent composition. The M5+, M6+, and M7+ rankings emphasise different processes—aftershock clustering for M5+, balanced evidence for M6+, long-term loading for M7+—through separate weight vectors, so the three maps are genuinely different orderings rather than rescalings of one field.

(D3) Explicit non-probability scoring. Display scores live on a bounded $[40, 99]$ scale produced by per-ranking min–max normalisation, and the product states plainly that they are relative ranks rather than occurrence probabilities.

Section 2 describes the catalogue and its growth. Section 3 states the full computational specification with worked examples. Section 4 presents the forecast product and its map figures. Section 5 defines the evaluation diagnostics, and Section 6 the limitations.

2 Data and Catalogue Construction

2.1 Source and shards

The source is the Sismik Harita earthquake-list API (`sismikharita.com/api`), which merges agency feeds (notably EMSC and TURHEC) with historical compilations. The repository bun-

dles the catalogue as immutable JSON shards: one accumulation shard for pre-2000 events and one shard per year for 2000–2026, totalling about 1.23×10^6 events in roughly 1.0 GiB. The 2026-08-01 forecast snapshot, for reference, was computed from 1,230,423 events with the newest event dated 2026-07-31T23:56:42Z. Each record carries an occurrence timestamp (UTC), latitude, longitude, magnitude, depth, a review label, the number of contributing agencies, the inter-source magnitude spread, and a primary-record flag. Records with unparseable or future timestamps, non-finite coordinates, or non-finite magnitudes are discarded, and the rest are clipped to the study rectangle.

2.2 Catalogue growth and completeness drift

Figure 1 plots the annual in-region event counts recovered from the 2000–2026 shards. Two facts stand out. First, yearly totals grew roughly tenfold, from 4,286 events in 2000 (17 at M5+) to 59,138 in 2025 (36 at M5+), reflecting denser networks and broader agency merging rather than a tenfold change in Earth behaviour. Second, individual years carry the imprint of productive sequences: 2011 (38,776 events), 2020 (43,247 events, 50 at M5+), and 2023 (92,214 events, 59 at M5+) each stand well above their neighbours because aftershock cascades contribute large numbers of locatable small events. Both facts motivate the rolling calibration design of Section 3.3: a fixed absolute time window would mix very different detection regimes, so completeness, b-value, and background rate are re-estimated on a trailing ten-year window at every evaluation.

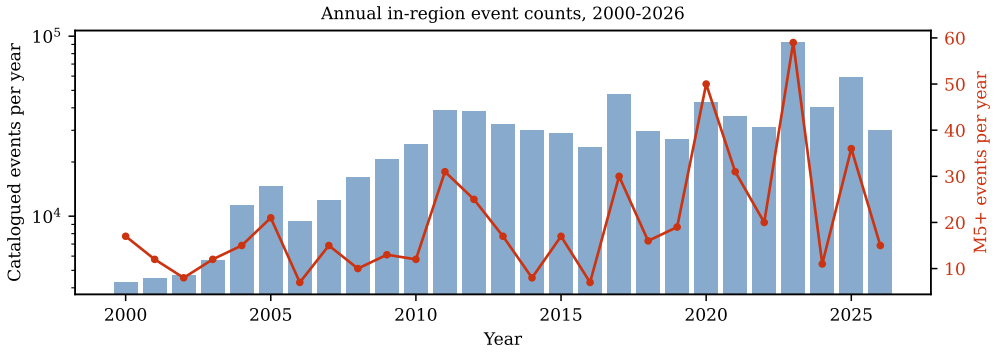


Figure 1: Annual in-region event counts from the bundled shards, 2000–2026 (left axis, logarithmic bars), with M5+ counts overlaid (right axis, line). The secular rise records improving detection; the spikes in 2011, 2020, and 2023 record highly productive sequences. The 2026 bar is a partial year.

2.3 Snapshot archive and calculation dates

A distinctive property of the system is that history is reproducible. Because the engine is deterministic and consumes only events at or before a chosen cut-off, the repository bundles 363 precomputed forecast snapshots (`data/snapshots/*.json`) covering millennial milestones, every year 1900–2026, and the elapsed months of the current year. A calculation-date selector maps each user-facing option to a bundle key: a month option `YYYY-MM` of the current year resolves to the first of that month; a standalone year 1900–2026 resolves to January 1 of that year; pre-1900 range labels resolve to their range-start year; and the current calendar day resolves to itself. The exclusive UTC cut-off for a key is the start of that date when the day field is 01 (so a year snapshot contains strictly the events before January 1) and the end of that date otherwise (so the current-day snapshot contains everything known). Serving a historical

selection therefore never recomputes: it reads the matching bundle, and any option missing from the archive is rejected rather than computed on demand.

2.4 Daily update pipeline

Once per Türkiye calendar day (UTC+3), at the first request after midnight, the server refreshes the catalogue from the provider starting 48 hours before the newest stored event. The overlap captures late arrivals and revisions; stale ranges are split into windows of at most 28 days; incoming records are normalised, de-duplicated by stable identity, and merged keeping the latest revision of each event. Forecast results are cached in memory, in temporary files, and in object storage under unique candidate names, with readers selecting the candidate of greatest catalogue coverage so that concurrent serverless instances cannot overwrite a complete result with a stale one.

2.5 Observation-quality weights

Every prepared event carries a quality weight $w \in [0.1, 1.0]$ that multiplies its contribution throughout the pipeline. The review label sets a base weight,

$$w_{\text{base}} = \begin{cases} 1.00 & \texttt{reviewed}, \texttt{focal_mechanism} \\ 0.90 & \texttt{turhec_extended} \\ 0.74 & \texttt{preliminary} \\ 0.62 & \texttt{automatic} \\ 0.48 & \texttt{historical_catalogue} \\ 0.82 & \text{otherwise,} \end{cases} \tag{1}$$

where `preliminary` denotes the provider's label for preliminary, unreviewed solutions. The final weight is

$$w = \text{clamp}\Big(w_{\text{base}} \cdot f_{\text{src}} \cdot f_{\text{spread}} \cdot f_{\text{depth}} \cdot f_{\text{primary}},\ 0.1,\ 1.0\Big), \tag{2}$$

with agency factor

$$f_{\text{src}} = \text{clamp}(0.94 + 0.03 \min(s_{\text{cnt}}, 3),\ 0.94,\ 1.03) \tag{3}$$

and spread penalty

$$f_{\text{spread}} = \text{clamp}(1.04 - 0.22\,\sigma_M,\ 0.68,\ 1.04) \tag{4}$$

applied when a spread is reported; $f_{\text{depth}} = 0.5$ for depths outside $[0, 700]$ km and 1 otherwise; and $f_{\text{primary}} = 0.5$ for non-primary duplicate records and 1 otherwise. Two worked examples fix the arithmetic. A reviewed event from two agencies with spread 0.2 receives $1.00 \times 1.00 \times 0.996 = 0.9960$. An automatic single-agency event with spread 0.5 receives $0.62 \times 0.97 \times 0.93 = 0.5593$. Implausible depths and duplicate records are therefore down-weighted but never silenced entirely, since even a doubtful record carries some information.

3 Methods

The engine is a pure function of (catalogue, reference timestamp). It runs once per Türkiye day to build an immutable base field, from which the full method×threshold×count matrix is derived without shared mutable state. What follows states each stage exactly, with the constants used by the implementation and worked numerical examples.

3.1 Grid geometry

The domain is a regular 0.5° grid with

$$\phi_{\min} = 34^\circ,\ \phi_{\max} = 43^\circ, \qquad \lambda_{\min} = 24^\circ,\ \lambda_{\max} = 46^\circ, \tag{5}$$

giving $N_\phi = \lceil 9/0.5 \rceil = 18$ rows, $N_\lambda = \lceil 22/0.5 \rceil = 44$ columns, and $N_{\text{cells}} = 792$ cells in row-major order. Row r is centred at $\bar{\phi}_r = 34 + (r + \frac{1}{2})0.5$ and column c at $\bar{\lambda}_c = 24 + (c + \frac{1}{2})0.5$. Cell area corrects for meridian convergence with $k = 111.195\,\text{km}\,\text{deg}^{-1}$:

$$A_{\text{cell}}(r) = (\Delta\phi \cdot k) \cdot (\Delta\lambda \cdot k \cdot \cos\bar{\phi}_r). \tag{6}$$

As an example, at $\bar{\phi} = 38.25^\circ$ the area is $55.5975^2 \times \cos 38.25^\circ = 3091.7 \times 0.78528 = 2427.48\,\text{km}^2$; across the domain the area ranges over roughly 2300–2700 km^2. Inner kernels use the flat-Earth distance $\mathrm{d}r = \sqrt{(\Delta\lambda\, k \cos\bar{\phi})^2 + (\Delta\phi\, k)^2}$, accurate at regional scales, while the recurrence geometry uses the exact haversine distance.

3.2 Event preparation

Raw records are parsed to UTC epoch seconds, rejected on invalid or future timestamps or non-finite coordinates or magnitudes, clipped to the rectangle, and assigned quality weights per Section 2.5, yielding prepared events $\{(\tau_i, \varphi_i, \lambda_i, M_i, w_i)\}$.

3.3 Magnitude of completeness

Calibration uses the trailing $W_{\text{cal}} = 10$ years of prepared events (the full catalogue only when fewer than 200 events remain), binned at $\Delta m = 0.1$ over 100 bins. Two estimators are combined. Maximum curvature (MAXC) takes the mode bin m^* of the non-cumulative distribution and adds a conservative half-bin correction, $M_{\text{MAXC}} = m^*\Delta m + 0.2$ [7]. Goodness-of-fit at 95 percent (GFT95) scans trial magnitudes 2.0–3.2 (requiring at least 200 events above trial) and accepts the smallest trial whose fitted G–R line reproduces cumulative counts within 5 percent relative absolute error:

$$1 - \frac{\sum_k \left|\Delta N_k^{\text{obs}} - \hat{N}(M_k)\right|}{\sum_k \Delta N_k^{\text{obs}}} \geq 0.95, \qquad \hat{N}(M_k) = N_{\geq M_{\text{trial}}} 10^{-b(M_k - M_{\text{trial}})}. \tag{7}$$

Completeness is the conservative envelope $M_c = \max(M_{\text{MAXC}}, M_{\text{GFT95}})$ clamped to $[2.0, 3.2]$, falling back to MAXC alone when GFT95 admits no candidate. A secondary Δa estimator after Godano and Petrillo scans trial thresholds and takes the first at which the G–R a-value jumps by more than $\min(\Delta a) + 0.3$; it is implemented for offline analysis.

Figure 2 applies these exact estimators to the 2016–2025 annual shards (430,859 in-region events). The non-cumulative mode bin is 13 (magnitude 1.3), so MAXC gives 1.5, which the floor clamp raises to 2.00; GFT95 independently returns 2.0. Hence $M_c = 2.00$ with $N_{\geq M_c} = 148{,}464$ and mean magnitude 1.81. The fitted G–R line in panel (b) uses the b-value of the next section. The lesson is instructive rather than alarming: a modern dense catalogue detects vast numbers of micro-earthquakes, and the clamp floor keeps the analysis safely above the detection roll-off.

3.4 Gutenberg–Richter layer

The global b-value is the Aki–Utsu maximum-likelihood estimate [5] over events with $M \geq M_c$,

$$b_{\text{global}} = \frac{\log_{10} e}{\langle M \rangle - (M_c - \Delta m/2)}, \quad \text{clamped to } [0.6, 1.4], \tag{8}$$

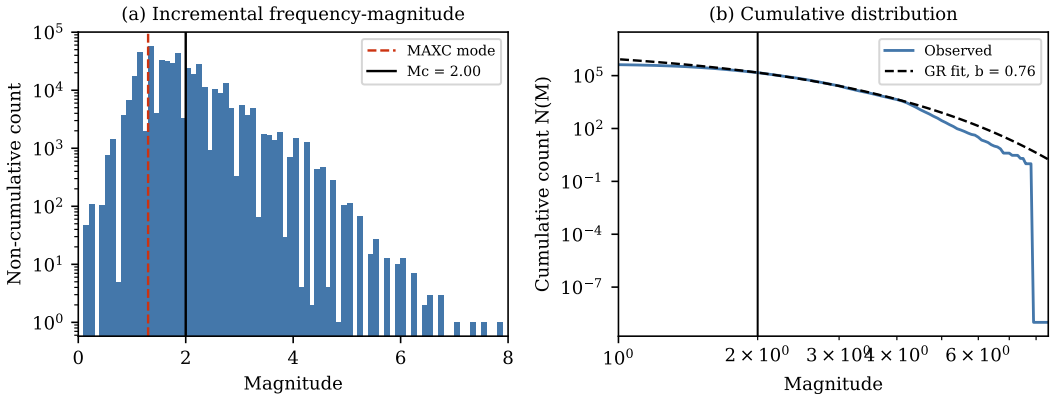


Figure 2: Frequency–magnitude distribution of the 2016–2025 in-region events (n = 430,859). (a) Non-cumulative counts per 0.1 bin with the MAXC mode (dashed) and the adopted M_c = 2.00 (solid). (b) Cumulative counts with the fitted G–R line at $b = 0.76$. Both estimators agree at the clamp floor.

with fallback 0.9 below 100 qualifying events. On the Figure 2 data, $\langle M \rangle = 1.8143$ is not directly comparable because the mean there spans all magnitudes; restricted to $M \geq 2.00$ the estimator yields $b = 0.7577$. The a-value $a = \log_{10} N_{\geq M_c} + bM_c$ gives $a = \log_{10}(148464) + 0.7577 \times 2.0 = 5.17157 + 1.51540 = 6.68702$, and the expected maximum magnitude $M_{\max}^{\mathrm{GR}} = a/b = 8.8254$, clipped to $[M_c + 0.3, 8.5]$ where an upper bound is required.

Spatially varying b-values pool all $M \geq M_c$ events in each cell's 3×3 neighbourhood ($1.5° \times 1.5°$) and apply Equation (8) with the global M_c; cells with fewer than 100 pooled events receive no local estimate. The non-negative anomaly $\Delta b = \max(0, b_{\mathrm{global}} - b_{\mathrm{local}})$ marks relatively stressed neighbourhoods [8, 9]. Rates at M_c are extrapolated to a target threshold by $S(M_{\mathrm{thr}}) = 10^{-b(M_{\mathrm{thr}} - M_c)}$ with the local b where available. For $b = 1.0$ and $M_c = 2.6$ the factors are $S(5) = 3.981072 \times 10^{-3}$, $S(6) = 3.981072 \times 10^{-4}$, and $S(7) = 3.981072 \times 10^{-5}$: each magnitude unit costs exactly one order of magnitude in rate, which is why the three thresholds rank different cells.

3.5 Declustering

The background field admits only independent events. Gardner–Knopoff (1974) windows [6] are applied in one chronological sweep with a rolling active-parent list:

$$\Delta t_{\mathrm{GK}}(M) = \begin{cases} 10^{\,0.5409\,M - 0.547}, & M < 6.5, \\ 10^{\,0.032\,M + 2.7389}, & M \geq 6.5, \end{cases} \quad \text{(days)}, \tag{9}$$

$$\Delta r_{\mathrm{GK}}(M) = 10^{\,0.1238\,M + 0.983} \quad \text{(km)}. \tag{10}$$

An event not larger than a still-active parent and inside *both* of its windows is flagged as dependent; survivors become parents in turn. Concrete window sizes: M5.0 spans 143.71 days and 39.99 km; M6.0 spans 499.34 days and 53.19 km; M6.5 spans 884.91 days and 61.33 km; M7.8 spans 973.87 days and 88.85 km. The kink at M6.5 is part of the published parameterisation, not an implementation choice. Stochastic EM declustering [13] is noted as a future replacement.

3.6 Background intensity

The background rate at each cell centre is the quality-weighted Gaussian density over declustered events, normalised by the complete span T_{cal} in days [10]:

$$\mu(\mathbf{x}_c) = \frac{A_{\text{cell}}(r)}{2\pi h^2}\,\frac{1}{T_{\text{cal}}}\sum_{j\in\mathcal{B}} w_j\,\exp\Big(-\frac{r_{jc}^2}{2h^2}\Big), \tag{11}$$

with $h = 25\,\text{km}$. The units are expected independent events per cell per day at M_c. A worked example: one unit-quality event at a cell centre with $A_{\text{cell}} = 2427.48\,\text{km}^2$ and $T_{\text{cal}} = 3650$ days contributes $2427.48/(2\pi\times625)/3650 = 1.693568\times10^{-4}$ per day. Scaled to M5+ with $b = 0.7577$ and $M_c = 2.0$, the factor is $10^{-0.7577\times3.0} = 5.332121\times10^{-3}$, giving an annual M5+ rate of 3.298254×10^{-4} whose base-ten logarithm is -3.4817—the quantity that enters the composite score of Section 3.12. A $3.5^\circ\times4.5^\circ$ bounding box keeps the $O(N_{\text{cells}}\times|\mathcal{B}|)$ evaluation tractable.

3.7 Triggered intensity

The triggered rate sums the separable ETAS kernel over all $M \geq M_c$ events at the reference time t,

$$\lambda_{\text{trig}}(\mathbf{x}_c, t) = \sum_{i\,:\,t_i<t} k(M_i)\;g(t-t_i)\;f\big(r_{ic}, M_i\big), \tag{12}$$

with the three factors below (Figure 3 shows their shapes).

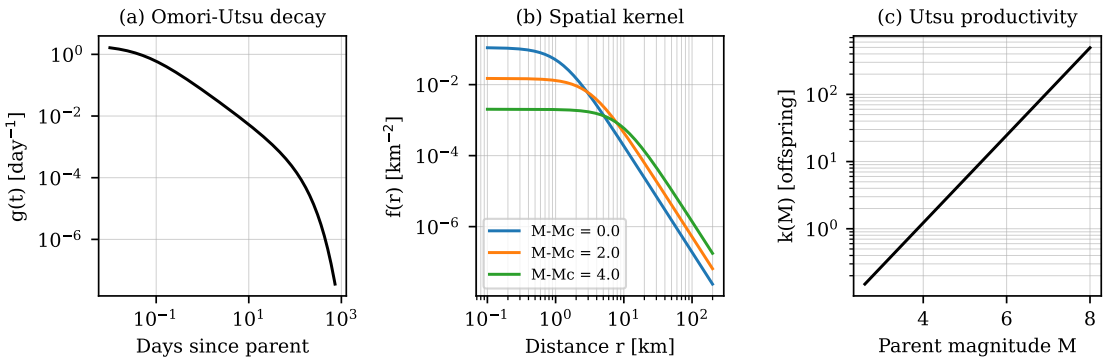


Figure 3: The three ETAS kernel factors at production parameters. (a) Omori–Utsu density $g(\Delta t)$ with $c = 0.05\,\text{d}$, $p = 1.1$, $\tau = 100\,\text{d}$. (b) Spatial density $f(r)$ for magnitude excess $M - M_c \in \{0, 2, 4\}$ above $M_c = 2.6$. (c) Utsu productivity $k(M)$ with $K = 0.15$, $\alpha = 1.5$.

Utsu productivity. The expected direct-offspring count is $k(M) = K\exp(\alpha(M - M_c))$ with $K = 0.15$ and $\alpha = 1.5$. At $M_c = 2.6$: $k(3.0) = 0.2733$, $k(5.0) = 5.4897$, $k(6.0) = 24.6033$, $k(7.0) = 110.2643$, $k(7.8) = 366.0903$. Because the form is exponential in the excess, ratios depend only on magnitude differences: $k(7)/k(5) = \exp(1.5\times2) = 20.0855$. An optional dual-productivity scheme (Zhang et al., 2020; disabled by default) assigns $\alpha_{\text{short}} = 2.0$ to the most recent $h\,10^{-bM_c}$ events ($h = 2\times10^5$) and $\alpha_{\text{long}} = 1.4$ to older ones.

Omori–Utsu decay. The tapered temporal density is

$$g(\Delta t) = \frac{p-1}{c}\Big(1+\frac{\Delta t}{c}\Big)^{-p}\exp\Big(-\frac{\Delta t}{\tau}\Big), \qquad c = 0.05\text{ d},\ p = 1.1,\ \tau = 100\text{ d}, \tag{13}$$

in days^{-1}, normalised to unit integral without the taper. Evaluated: $g(0) = 2.0$, $g(0.5) = 0.1423399$, $g(1) = 0.06954178$, $g(7) = 0.008062846$, $g(30) = 0.001300109$, $g(365) = 2.925058\times10^{-6}$, $g(730) = 3.546986\times10^{-8}$. The taper removes the $t\to0$ singularity from numerical reach and imposes finite memory; residual mass beyond 730 days is about 6.2×10^{-6}, which justifies truncating the sum there. Combined with productivity, an M6.0 parent contributes a space-integrated $24.6033\times0.06954178 = 1.710957$ events per day one day after the shock, falling to $24.6033\times0.001300109 = 0.031987$ per day after thirty days.

Spatial kernel. Offspring spread as the power law

$$f(r, M) = \frac{q-1}{\pi\, d(M)^2}\left(1 + \frac{r^2}{d(M)^2}\right)^{-q}, \qquad q = 1.5, \tag{14}$$

$$d(M) = D_0\, \exp\big(\gamma\,(M - M_c)\big), \qquad D_0 = 1.2 \text{ km},\ \gamma = 0.5, \tag{15}$$

with $d(M)$ clamped to $[1, 400]$ km and unit integral over $\mathbb{R}^2$. At $M = M_c$ ($d = 1.2$ km): $f(0) = 1.105243 \times 10^{-1}$, $f(10\,\text{km}) = 1.869337 \times 10^{-4}$, $f(50\,\text{km}) = 1.526568 \times 10^{-6}$ per km^2. At excess 2.0 ($d = 3.2619$ km): 1.495783×10^{-2}, 4.460938×10^{-4}, 4.126854×10^{-6}. At excess 4.0 ($d = 8.8669$ km): 2.024323×10^{-3}, 5.911400×10^{-4}, 1.077728×10^{-5}. Per-cell weights multiply by cell area, clamp to 1, and renormalise across touched cells per parent, so narrow kernels keep their mass on a coarse grid.

3.8 Threshold scaling

For $M_{\text{thr}} \in \{5, 6, 7\}$ the total daily rate is

$$\lambda_{\text{total}}(M_{\text{thr}}) = \big[\mu(\mathbf{x}_c) + \lambda_{\text{trig}}(\mathbf{x}_c, t)\big]\, 10^{-b\,(M_{\text{thr}} - M_c)}, \tag{16}$$

annualised by 365.2425. The clustering ratio is

$$\rho = \frac{\lambda_{\text{trig}}}{\mu + \lambda_{\text{trig}}}, \tag{17}$$

which diagnoses how much of a cell's rate is triggered rather than background.

3.9 Seismicity indicators

Indicators use a trailing 5-year window (energy, CV, natural time, rate change); M_c, b, and μ use 10 years. Accumulation is per cell, then pooled over 3×3 neighbourhoods.

Coefficient of variation. Over inter-event gaps, requiring at least 3 gaps [18, 19],

$$\text{CV} = \frac{\sigma_{\Delta t}}{\mu_{\Delta t}}, \qquad \sigma^2_{\Delta t} = \frac{\sum_k \delta_k^2 - (\sum_k \delta_k)^2/n}{n-1}. \tag{18}$$

Example: gaps $[10, 12, 9, 40]$ days give mean 17.75, sample deviation 14.89, and CV = 0.8386 (near-periodic); gaps $[1, 2, 1, 60]$ give mean 16.0, deviation 29.34, and CV = 1.8336 (strongly clustered).

Root-energy rate. With $\log_{10} E = 1.5M + 4.8$ [11], $\sqrt{E_i} = 10^{0.75 M_i + 2.4}$ and $\mathrm{d}E^{1/2}/\,\mathrm{d}t = T_{\text{ind}}^{-1} \sum_i \sqrt{E_i}$ in J$^{1/2}$ yr^{-1}. Reference values: M3.0 releases 1.9953×10^9 J ($\sqrt{E} = 4.4668 \times 10^4$); M5.0 releases 1.9953×10^{12} J (1.4125×10^6); M6.0 releases 6.3096×10^{13} J (7.9433×10^6); M7.0 releases 1.9953×10^{15} J (4.4668×10^7). The indicator measures released energy, never stored energy.

Natural-time progress. With $M_S = 3.0$ and $M_T = 5.0$ [16, 17], $N_{\text{GR}} = 10^{b(M_T - M_S)}$ and $\pi_{\text{nt}} = 1 - \exp(-n/N_{\text{GR}})$, where n counts small events since the last local target reset. For $b = 1.0$, $N_{\text{GR}} = 100$, and $n = 0, 20, 50, 100, 200, 400$ give progress 0.000000, 0.181269, 0.393469, 0.632121, 0.864665, 0.981684. Cells without any past target event receive no value. This is a G–R-normalised proxy, not an empirical probability.

Rate-change z. Habermann's statistic [20] with $T_{\text{rec}} = 365$ days,

$$z = \frac{r_{\text{rec}} - r_{\text{long}}}{\sqrt{r_{\text{long}}/T_{\text{rec}}}}, \qquad r_{\text{rec}} = \frac{N_{\text{rec}}}{T_{\text{rec}}}, \quad r_{\text{long}} = \frac{N_{\text{long}}}{T_{\text{ind}}}. \tag{19}$$

Example: 12 events in the recent year against 80 in five years gives $r_{\text{rec}} = 0.032877$, $r_{\text{long}} = 0.043806$, and $z = -0.9981$, mild quiescence. The value is reported but excluded from the composite score after quarterly checks showed no stable skill for the next 90-day M5+ outcome.

Magnitude deficit. $\Delta M = M_{\text{max}}^{\text{GR}} - M_{\text{max}}^{\text{obs}}$ clamped to $[-1, 5]$, diagnostic only. On the Figure 2 numbers, $M_{\text{max}}^{\text{GR}} = 8.8254$, so a cell whose observed maximum is 7.8 reports $\Delta M = 1.0254$.

3.10 Large-event recurrence

Mainshocks with $M \geq 6$ are separately declustered, sequence-collapsed (minimum gaps 2 years for M6+, 3 years for M7+), and collected within 75 km (M6+) or 110 km (M7+) neighbourhoods; fewer than three independent mainshocks yields no value rather than zero. Sample intervals give mean $\bar{\delta}$ and aperiodicity shrunk toward the prior 0.6 with prior size 3, clamped to $[0.35, 1]$. The Brownian Passage Time density, CDF, and 30-year conditional probability are

$$f_{\mathrm{BPT}}(t;\mu,\alpha) = \sqrt{\frac{\mu}{2\pi\alpha^2 t^3}}\,\exp\Big[-\frac{(t-\mu)^2}{2\mu\alpha^2 t}\Big], \tag{20}$$

$$F_{\mathrm{BPT}}(t) = \Phi(u_1) + e^{2/\alpha^2}\,\Phi(-u_2), \qquad u_{1,2} = \frac{\sqrt{t/\mu} \mp \sqrt{\mu/t}}{\alpha}, \tag{21}$$

$$P_{30}^{\mathrm{BPT}} = \frac{F_{\mathrm{BPT}}(t_e+30) - F_{\mathrm{BPT}}(t_e)}{1 - F_{\mathrm{BPT}}(t_e)}, \tag{22}$$

with Φ by the Abramowitz–Stegun approximation (error below 7.5×10^{-8}) [14, 15]. Worked example with mean recurrence 100 years, aperiodicity 0.6, and 80 elapsed years: $F(80) = 0.457689$ and $F(110) = 0.672629$, so that

$$P_{30}^{\mathrm{BPT}} = \frac{0.672629 - 0.457689}{1 - 0.457689} = 0.396341, \tag{23}$$

against a Poisson baseline $1 - e^{-30/100} = 0.259182$. The renewal memory thus raises the conditional probability by more than half relative to the memoryless model at this elapsed fraction. The reported probability blends BPT with Poisson by confidence $\kappa = \mathrm{clamp}(n_{\mathrm{int}}/6, 0, 1)\cdot(0.5 + 0.5\bar{w})$, $P_{30} = \kappa P_{30}^{\mathrm{BPT}} + (1-\kappa)[1 - e^{-30/\bar{\delta}}]$, scored as $P_{30}(0.5 + 0.5\kappa)$.

3.11 Centroids

Reporting positions are decay-weighted mean epicentres with weights

$$w_{\mathrm{cen}} = w_i \cdot 10^{0.3(M_i - M_c)} \cdot e^{-\,\mathrm{age}_i/15\ \mathrm{yr}}. \tag{24}$$

Example: a unit-quality M6.0 event at $M_c = 2.6$ seen two years ago carries $10^{1.02} \times e^{-2/15} = 10.471 \times 0.8752 = 9.1642$, so large recent events dominate the centroid while old small ones fade.

3.12 Composite score

The cell score blends the log-annual rate with bounded indicator terms and a magnitude-feasibility term:

$$\begin{aligned} H_{\mathrm{cell}} = {} & \log_{10}\big(\lambda^{\mathrm{ann}} + \varepsilon\big) + \ln\big[f_{\mathrm{feas}}(M_{\mathrm{thr}}, M_{\max}^{\mathrm{obs}})\big] \\ & + w_b\ \mathrm{clamp}\Big(\frac{\Delta b}{b_{\mathrm{global}}}, 0, 0.5\Big) + w_{\mathrm{nt}}\ \mathrm{clamp}\big(\pi_{\mathrm{nt}} - 0.5,\ -0.5,\ 0.5\big) \\ & + w_{\mathrm{en}}\ \tanh\Big(\frac{\mathrm{d}E^{1/2}/\,\mathrm{d}t}{10^9}\Big) + w_{\mathrm{cv}}\ \mathrm{clamp}\Big(\frac{\mathrm{CV}-1}{2},\ 0,\ 1\Big) \\ & + w_{\mathrm{rec}}\,P_{30}\,\big(0.5 + 0.5\kappa\big), \end{aligned} \tag{25}$$

with $\varepsilon = 10^{-12}$ and

$$f_{\mathrm{feas}}(M_{\mathrm{thr}}, M_{\max}) = \exp\Big[-\frac{\max(0, M_{\mathrm{thr}} - M_{\max})^2}{2\cdot(0.8)^2}\Big]. \tag{26}$$

Feasibility is a soft Gaussian penalty, never a hard cut, because a finite catalogue cannot prove an unobserved magnitude impossible. For a cell with $M_{\max}^{\text{obs}} = 5.3$: $f_{\text{feas}}(5) = 1.000000$ ($\ln = 0.0000$), $f_{\text{feas}}(6) = 0.681941$ ($\ln = -0.3828$), $f_{\text{feas}}(7) = 0.104579$ ($\ln = -2.2578$). The M7+ ranking therefore discounts such a cell by 2.26 score units before any indicator is considered, which is the main reason the three thresholds order cells differently. Every modifier is bounded, so no outlier can dominate.

Table 1: Composite-score weights by magnitude threshold.

Weight	M5+	M6+	M7+	Dominant rationale
w_b (b-value anomaly)	0.03	0.07	0.14	long-term stress accumulation
w_{nt} (natural time)	0.05	0.09	0.14	long-cycle progress
w_{en} (energy rate)	0.04	0.06	0.10	released-energy activity
w_{cv} (clustering)	0.12	0.06	0.02	aftershock clustering (M5+)
w_{rec} (recurrence)	0.00	0.50	1.00	long-term repeat history

3.13 Selection and normalisation

Candidates sort by composite score and survive greedily only when at least 1.8 cell widths (about 100 km) from every accepted cell, so one aftershock zone cannot occupy the whole list. The 50 survivors are min–max normalised,

$$s_{\text{disp}} = 40 + 59 \cdot \frac{H_{\text{cell}} - H_{\min}}{H_{\max} - H_{\min}} \in [40, 99], \tag{27}$$

and banded at ≥ 92 (very high), ≥ 80 (high), ≥ 65 (notable), and below (elevated). Example: with pool extremes -6.2 and -2.1, a cell at -3.0 maps to $40 + 59 \times 3.2/4.1 = 86.0488$, reported as 86.0 (high). The display score is a within-ranking position, not a probability.

3.14 Single-signal methods

The same field yields eight single-signal orderings: Poisson ($\log_{10} \mu^{\text{ann}}$), ETAS ($\log_{10} \lambda^{\text{ann}}$), Triggered, b-value anomaly, natural time, energy, clustering, and historical recurrence. Each is normalised independently and scores cannot be compared across methods. The b-value, natural-time, energy, and clustering layers are threshold-independent by construction.

3.15 Parameter inventory

Table 2 collects every numerical constant for reproducibility in one place; all symbols are defined in the sections above.

3.16 Computational implementation

Fields are `Float64Array` buffers of length 792; per-event characteristic radii and elapsed days are cached before the inner cell loop; cosines of row latitudes are precomputed once; and both kernel summations apply the $3.5^\circ \times 4.5^\circ$ bounding box before any transcendental evaluation. The background pass costs $O(N_{\text{cells}} \times |\mathcal{B}|)$ and the triggered pass $O(N_{\text{cells}} \times E_{730})$ with E_{730} the events of the last 730 days, after which neighbourhood pooling is linear in the cell count. No stage shares mutable state across thresholds or methods, and identical inputs reproduce identical outputs bit-for-bit subject only to floating-point determinism of a single thread.

Table 2: Complete numerical parameter inventory of the engine.

Symbol	Value	Role
$\Delta\phi$	0.5°	grid cell size
Domain	34–43°N, 24–46°E	study rectangle
N_{cells}	$18 \times 44 = 792$	grid size
k	$111.195\,\text{km}\,\text{deg}^{-1}$	degree-to-km factor
W_{cal} / W_{ind}	10 / 5 years	calibration / indicator windows
Δm, bins	0.1, 100	FMD binning
M_c clamp	$[2.0, 3.2]$, GFT 95 percent	completeness envelope
b clamp, $N^b_{\min}$	$[0.6, 1.4]$, 100 events	G–R slope guard
$M_{\max}$ clip	$[M_c + 0.3, 8.5]$	upper-bound guard
h	25 km	background bandwidth
K, α	0.15, 1.5	Utsu productivity
c, p, τ	0.05 d, 1.1, 100 d	Omori–Utsu decay
Lookback	730 days	trigger summation cap
q, D_0, γ	1.5, 1.2 km, 0.5	spatial kernel
$d(M)$ clamp	$[1, 400]$ km	aftershock-zone guard
T_{rec}	365 days	rate-change recent window
CV gaps	≥ 3	variation guard
M_S, M_T	3.0, 5.0	natural-time counting levels
ΔM clamp	$[-1, 5]$	deficit guard
Recurrence radii	75 km (M6+), 110 km (M7+)	neighbourhoods
Sequence gaps	2 yr (M6+), 3 yr (M7+)	collapse rule
α_0, clamp	0.6 prior, $[0.35, 1]$	aperiodicity guard
H	30 years	recurrence horizon
Feasibility σ	0.8	magnitude-support width
Separation	1.8 cells	spatial de-duplication
Display range	$[40, 99]$, pool 50	normalisation
Radii	75 / 120 / 180 km	M5+ / M6+ / M7+ influence

4 Forecast Product

The dashboard renders the top-ranked signals on an interactive Leaflet map (Figure 4) with a method selector, a magnitude-threshold selector, a signal-count selector, and the calculation-date selector of Section 2. Each marker reports its rank, centroid, influence radius, display score, signal band, and the full indicator suite, including annualised background, triggered, and total rates, the clustering ratio, b-value and completeness, quality-weighted counts, recency, energy rate, CV, natural-time progress, rate-change z, magnitude deficit, and the recurrence quadruple where defined.

Figure 5 plots the archived combined-M7+ top 50 of the 2026-08-01 snapshot over the model grid. The two highest cells score 99 at $(34.82^\circ\text{N}, 25.30^\circ\text{E})$ in the Hellenic arc southwest of Crete and at $(34.97^\circ\text{N}, 45.09^\circ\text{E})$ on the eastern edge, followed by 85.1 at $(35.36^\circ\text{N}, 27.15^\circ\text{E})$ and 84.8 at $(36.34^\circ\text{N}, 25.90^\circ\text{E})$; the schematic NAFZ and EAFZ traces are drawn for orientation only. Figure 6 places the M5+, M6+, and M7+ combined fields side by side from the same snapshot: the orderings visibly differ, with M5+ emphasising Aegean and western Anatolian clustering and M7+ concentrating on the Hellenic arc and the eastern borderlands, exactly as the weight schedule of Table 1 intends.

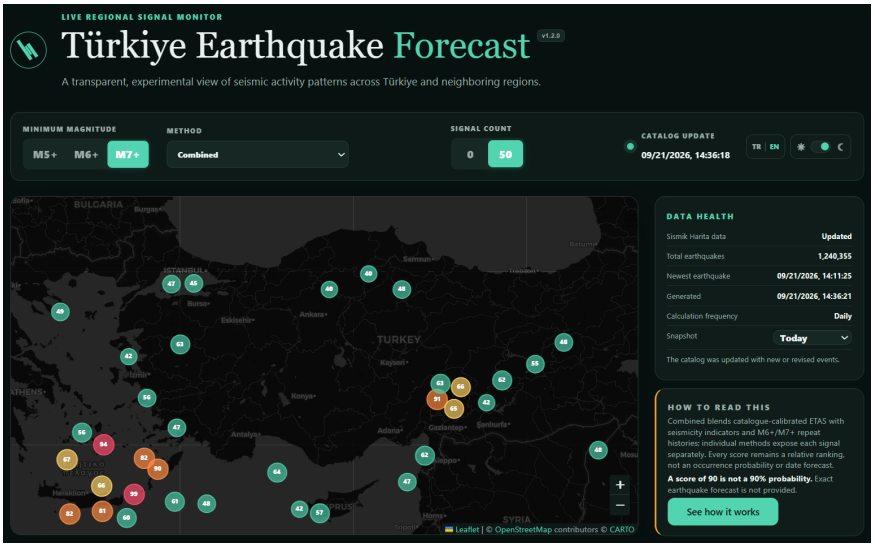


Figure 4: The interactive dashboard rendering the ranked signal field on a Leaflet map. Marker size and colour encode the display score; the calculation-date selector reproduces any archived historical field.

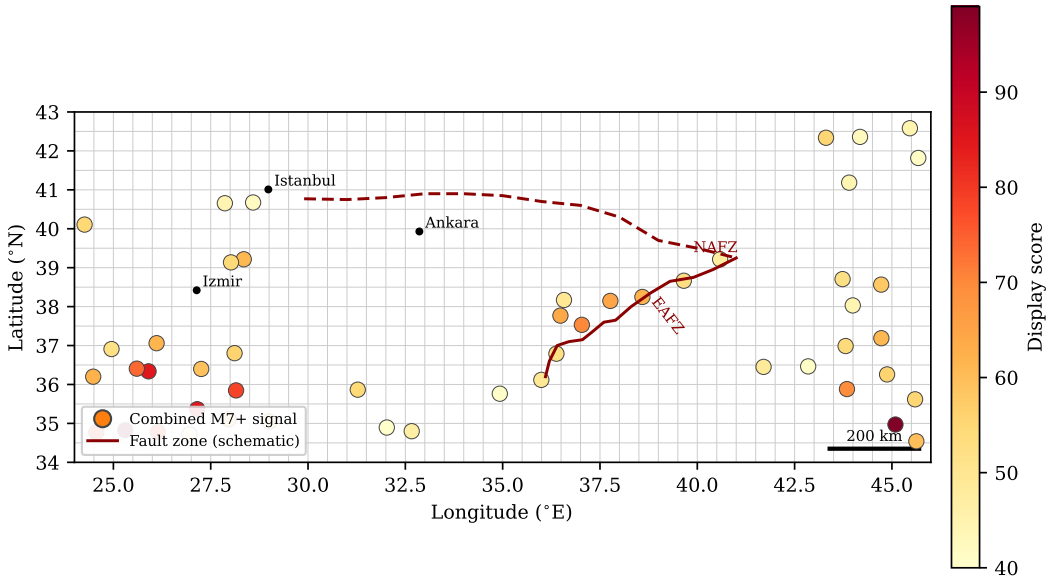


Figure 5: Combined-M7+ top-50 ranking field from the 2026-08-01 snapshot (catalogue cut-off 2026-07-31, 1,230,423 events). Circle colour encodes the display score on $[40, 99]$; fault traces are schematic and shown for orientation; the 200 km bar is computed at 38.5°N.

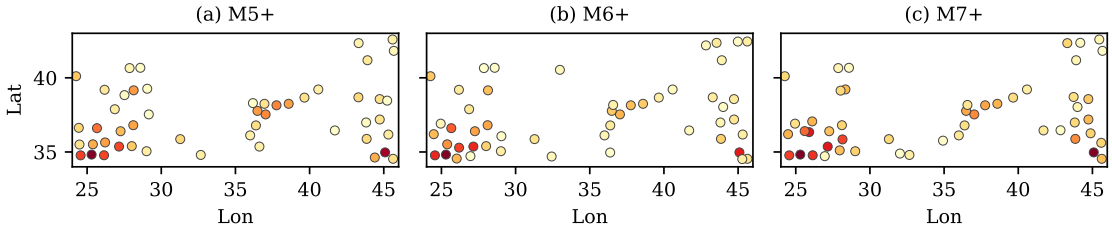


Figure 6: Combined top-50 signals by target threshold from the same 2026-08-01 snapshot. The three orderings differ because feasibility, clustering, and recurrence weights are threshold-dependent (Table 1).

5 Evaluation and Diagnostics

Offline testing uses CSEP-style primitives. The Poisson log-likelihood drops the constant factorial term,

$$\mathcal{L} = \sum_i \left[y_i \log \lambda_i - \lambda_i \right], \tag{28}$$

the information gain per event is

$$\mathrm{IGPE} = \frac{\mathcal{L}_{\mathrm{target}} - \mathcal{L}_{\mathrm{ref}}}{N_{\mathrm{obs}}}, \tag{29}$$

and the Brier score with its skill score is

$$\mathrm{BS} = \frac{1}{N} \sum_i (f_i - o_i)^2, \qquad \mathrm{BSS} = 1 - \frac{\mathrm{BS}}{\mathrm{BS}_{\mathrm{ref}}}. \tag{30}$$

The monotone Poisson transform $P = 1 - e^{-\lambda H}$ applies for fixed horizons. The accompanying software studies report three retrospective findings that shaped the production configuration. First, a chronological 2012–2025 pseudo-prospective comparison of calibration windows found the 10-year background at mean M5+ spatial information gain 0.658 per event over 2019–2025 against 0.442 for 20 years, with 3-, 5-, and 10-year indicator windows scoring 0.51126, 0.51115, and 0.51001; the differences among indicator windows are negligible and five years is retained as the stable middle choice. Second, chronological 30-year holdouts of the recurrence layer reached mean cell-level AUC 0.697 for M6+ and 0.596 for M7+, weak-to-moderate discrimination that justifies the confidence blending and subordinate role of Section 3.10. Third, quarterly pseudo-prospective checks of the rate-change z-value found no stable discrimination for the following 90-day M5+ outcome, which is why z and the magnitude deficit are reported but unscored. All figures are retrospective and await prospective confirmation.

6 Discussion and Limitations

The engine is conservative by construction, and four limitations from the earlier specification stand, joined here by three further qualifications.

First, completeness drift (Section 2) is mitigated by rolling estimation but never removed; pre-1900 entries are sparse and detection has improved tenfold within the instrumental era.

Second, Gardner–Knopoff declustering is a deterministic heuristic; the background rate inherits its errors, and stochastic alternatives [13] or time-varying completeness corrections [26] would improve μ.

Third, the composite weights are fixed engineering values rather than likelihood fits, so no occurrence probability is, or should be, derived from the display score.

Fourth, nothing beyond the catalogue enters: fault geometry, slip rates, geodetic locking, and stress transfer are absent, so the M7+ layer states catalogue regularity, not physical constraint.

Fifth, the grid is coarse relative to rupture dimensions. An M7+ rupture spans tens of kilometres [12], comparable to a 0.5° cell, so near-field ranking cannot resolve along-strike variations; the 1.8-cell separation and the 75–180 km influence radii are explicit acknowledgements of this resolution limit.

Sixth, all kernels are isotropic and depth-blind: directivity, fault orientation, and the seismogenic thickness play no role, which flatters circular aftershock zones over elongated ones.

Seventh, the daily cadence and the 730-day trigger cap mean the model responds to sequences within days but forgets them within years by design; it is a ranking of the present state, not a memory of all history.

The standing disclaimer follows from these points: relative ranks, not probabilities, and no use as a warning system.

7 Conclusion

This paper has stated, equation by equation and example by example, a reproducible four-layer catalogue-only engine for ranking regional seismic signals in Türkiye and neighbouring regions at M5+, M6+, and M7+ thresholds. Full ETAS spatio-temporal modelling, Gutenberg–Richter completeness and b-value mapping, six bounded indicators, and a Brownian-Passage-Time recurrence layer are composed with threshold-dependent weights and normalised to a non-probabilistic display scale; the parameter inventory, the worked examples, and the six figures fix the behaviour completely. Prospective testing of the rankings against future events is the natural next step.

Code and Data Availability

The engine is open source under the MIT license. Repository:

`https://github.com/bariskisir/TurkiyeEarthquakeForecast`

Live dashboard:

`https://turkiye-earthquake-forecast.vercel.app`

Bundled catalogue shards and the 363 precomputed forecast snapshots ship with the repository; daily updates are retrieved from the Sismik Harita API (`sismikharita.com/api`).

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